

EPAM — Round Two Question Bank

Every question asked in the round, as reported. Items in **blue** were flagged as weak or unanswered. Nothing here is a prepared answer — this is the raw question set to work through.

1 Live Coding

1. **Given a list of comma-joined strings, produce a descending list of distinct numbers, ignoring any non-numeric token.**

Not completed in the round. Highest-priority practice item.

```
List<String> strArr = Arrays.asList("1,2,3", "3,4,5", "6,abc,7",  
                                   "8,9,10", "10,2,4");  
  
// Expected output: [10, 9, 8, 7, 6, 5, 4, 3, 2, 1]
```

Behaviours the problem hides: split and flatten; skip "abc" without throwing; de-duplicate (3, 10, 2 and 4 repeat); sort descending; sort *numerically*, not lexicographically — otherwise 10 lands beside 1.

2. A second coding problem was also given.

Completed successfully in the round.

2 Java Core — JVM and Memory

3. Explain the JVM memory model and its regions.
4. Stack memory versus heap memory.
5. How does garbage collection work?
6. **In Java 8, which garbage collector is used?**
7. What is the String pool, and why is it needed? Cover the memory implications.
8. String is immutable — how is that beneficial, or is it not?
9. Can the heap size be increased? Where is that done — application level or DevOps level?
10. **How would you migrate a service from Java 8 to Java 11? Where do the changes go, and how would you plan it?**

Answered only up to "the JAR builds" — roughly 20% of the real work. Reviewed in follow-up session.

3 Collections Internals

11. ArrayList versus LinkedList.
12. How does HashMap work internally?
13. How does LinkedHashMap work internally?
14. **How is a collision handled when data is added?**
15. What is the default size of an ArrayList, and of the other collection classes?

4 Streams and Functional Programming

16. **What are the disadvantages of streams?**

Asked as a negative. Most preparation only covers the benefits.

17. Streams versus grep.

18. **How is grep coded internally?**

19. If I am reading a stream in parallel, how do I stop further processing once I have what I want?

20. What is the difference between object-oriented programming and functional programming?

5 Spring and Spring Boot

21. Spring basics.

Described as being asked at a very basic level.

22. **If I write @Service on a Repository class, what is the issue — and will Spring even identify it?**

23. **Lazy initialization in JPA when fetching from the database. What if I want full data for some users and partial data for others?**

24. Spring Security — JWT versus session.

25. What is Swagger, why does it exist, and what is it used for?

26. Which HTTP methods are idempotent?

27. Microservice design patterns.

28. Explain the difference between monolithic and microservice architecture using a health application as the example.

6 Architecture and Distributed Systems

29. Which AWS services have you used? EC2, storage, load balancer — just name them.

30. **Have you ever encountered a scenario where a CDN served incorrect image data?**

Graded on whether you have actually hit this in production, not on the definition.

31. **CAP theorem — when data is written and read at the same time, how do you ensure the reader does not get incorrect data?**

32. **If I copy your JWT — say a JioHotstar token — can someone else watch the content for free? How would you handle this?**

A genuinely good question. A bearer token is bearer: possession is authorisation. The fix is not inside the token.

7 Testing

33. Explain the testing pyramid model.

34. Explain the test-driven development approach.

8 Operations and Debugging

35. **How do you debug an issue using a log file containing more than a million entries?**

Still to capture: further questions were asked, mostly on the Spring side. Add them to this list as they are recalled.

Pattern in this round: roughly half the questions are pure recall — string pool, default ArrayList size, which garbage collector. Finite, memorisable, a few focused hours of work. The other half are applied scenarios — the CDN image, the shared token, partial versus full JPA fetch, a million log lines. Those carry the senior signal and are graded on whether you have met the problem in production. Neither half is system-design depth, which is where the preparation went.